Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route ospf 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129 2

Does R2 show up as an OSPF neighbor on R1? __Yes_ 2

Does R2 show up as an OSPF neighbor on R3? _No__ 2

What does this information tell you?

R2 and R3 cannot establish an OSPF neighbor adjacency directly, and R3 must reach the 192.168.2.0/24 network indirectly through R1. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___ S0/0/1___ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

2 _______65. R3 → R2 serial link (T1 1.544 Mb/s) = 64; R2 G0/0 LAN link (Gigabit Ethernet) = 1

Is R2 listed as an OSPF neighbor to R3? ___Yes__ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The modern networks often have links faster than this 100 Mb/s. Increasing the reference bandwidth allows OSPF to calculate more accurate and differentiated costs for high-speed links.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24 network: 782

Path: R1 → R3 → 192.168.3.0/24 ; R1 S0/0/1 (serial, bandwidth 128 Kbps): Cost = 100,000 / 128 = 781

2

R3 G0/0 (Gigabit Ethernet): Cost = 1;

Cost to 192.168.23.0/30 network: 845

Path: R1 → R3 → 192.168.23.0/30; R1 S0/0/1 (serial, bandwidth 128 Kbps): Cost = 100,000 / 128 = 781

R3 S0/0/1 (serial, default 1544 Kbps): Cost = 100,000 / 1544 $\approx$ 64

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562; After applying bandwidth 128 to all serial interfaces, each T1 serial link now has an OSPF cost of 781 (100,000 / 128 $\approx$ 781). Since the path to the 192.168.23.0/24 network traverses two serial links (R1 → R2/R3 → destination), the accumulated cost is: 781 (first serial link) + 781 (second serial link) = 1,562

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF always selects the path with the lowest accumulated cost.

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- Path via R2: R1→R2→R3 = 781 + 781 + 1 = 1,563
- Direct path via R3: R1→R3 = 1,565 + 1 = 1,566

Since 1,563 < 1,566, OSPF prefers the route through R2. The `ip ospf cost 1565` command manually increased the cost on R1's S0/0/1 interface, making the direct path less favorable.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   A stable Router ID ensures consistent OSPF operation. If the Router ID is automatically derived from a physical interface IP and that interface goes down, the Router ID may change—causing OSPF neighbor adjacencies to reset and triggering unnecessary SPF recalculations. Using a loopback interface or the router-id command guarantees a permanent, predictable Router ID.

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2. Why is the DR/BDR election process not a concern in this lab?

   DR/BDR elections only occur on multi-access broadcast networks (like Ethernet). This lab uses only point-to-point serial links, where OSPF neighbors form adjacencies directly and no DR/BDR election is needed.

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3. Why would you want to set an OSPF interface to passive?

   Setting an interface to passive prevents OSPF Hello packets from being sent out that interface. This is useful for LAN interfaces connected to end devices (like PCs) that don't run OSPF because it reduces unnecessary traffic and improves security, while still allowing the router to advertise the connected network to other OSPF neighbors.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF uses a multi-area hierarchical design where Area 0 serves as the backbone connecting all other areas. This structure limits the scope of link-state updates and SPF calculations to individual areas, reducing resource usage and containing topology changes, which enables large scalable networks.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   ________________ network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route will be used. When multiple routing protocols provide routes to the same destination, the router selects based on Administrative Distance, not metric. EIGRP has an AD of 90, OSPF has 110, and RIP has 120. Since lower AD is preferred, EIGRP wins regardless of its higher metric value.

   4